

Problem 2.8

Suppose

$$a(t) = at^2 + 10b.$$

If \$X invested at time 0 accumulates to \$1,000 at time 10, and to \$2,000 at time 20, find the original amount of the investment X .

Solution

Since the amount function is

$$a(t) = at^2 + 10b,$$

we use the two given accumulated values.

At $t = 10$,

$$a(10) = a(10)^2 + 10b = 100a + 10b = 1000.$$

At $t = 20$,

$$a(20) = a(20)^2 + 10b = 400a + 10b = 2000.$$

Subtracting the first equation from the second,

$$(400a + 10b) - (100a + 10b) = 2000 - 1000,$$

so

$$300a = 1000.$$

Therefore,

$$a = \frac{10}{3}.$$

Substitute this into the first equation:

$$100 \left(\frac{10}{3} \right) + 10b = 1000.$$

Thus,

$$\frac{1000}{3} + 10b = 1000,$$

so

$$10b = 1000 - \frac{1000}{3} = \frac{2000}{3}.$$

Hence,

$$b = \frac{200}{3}.$$

Now evaluate the original amount at time $t = 0$:

$$X = a(0) = a(0)^2 + 10b.$$

Since $0^2 = 0$,

$$X = 0 + 10 \left(\frac{200}{3} \right) = \frac{2000}{3}.$$

Therefore,

$$X = \frac{2000}{3} \approx 666.67.$$